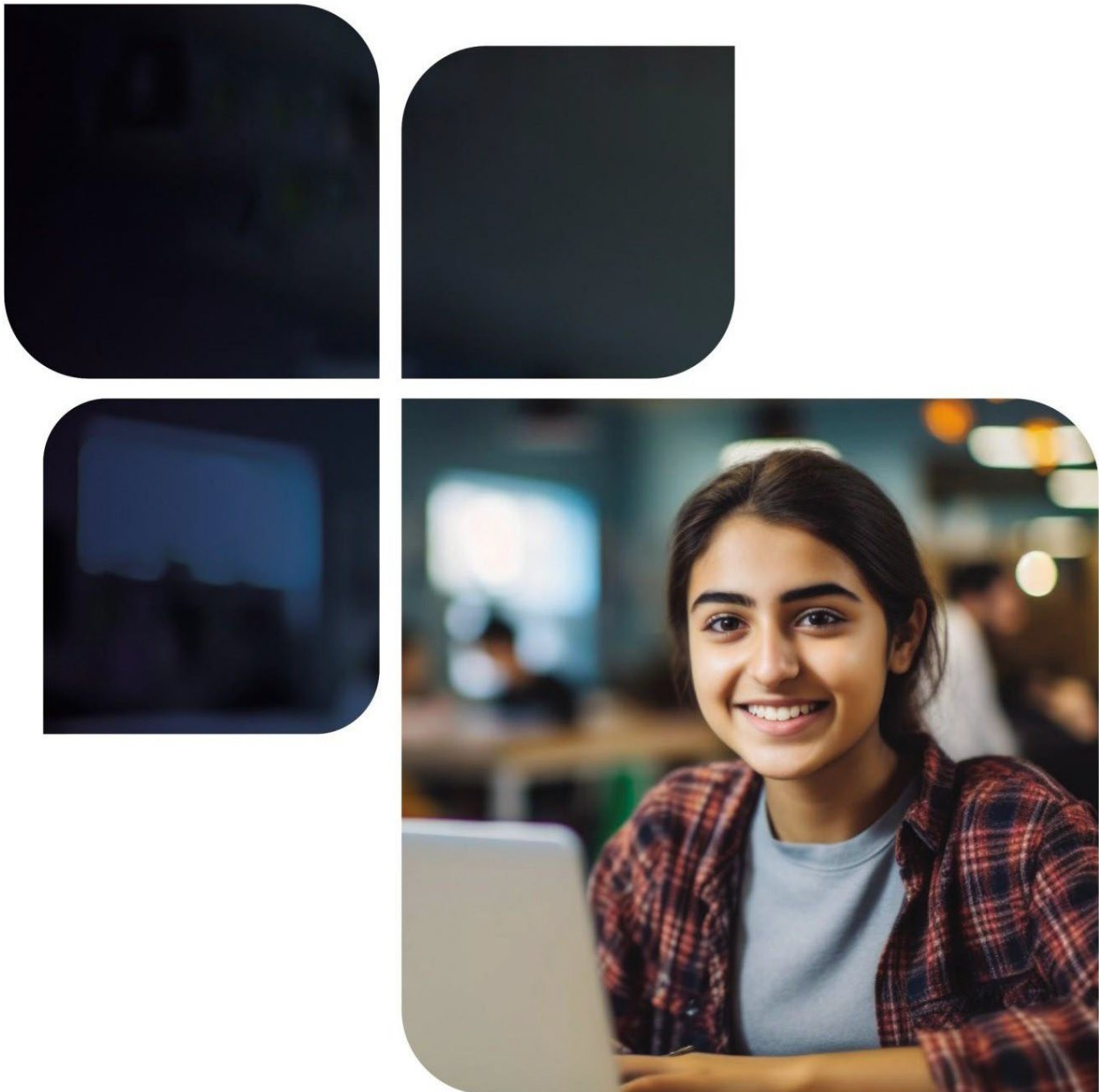


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Billing System Project Report

A report Submitted in fulfillment of the requirements for the organization



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Abstract

The Billing System Project is developed using Python programming language with Object-Oriented Programming concepts. This project is designed to simplify the billing process in shops and supermarkets. The system allows users to view available products along with their prices. Customers can select products and enter the required quantity for purchase. The program automatically calculates the total amount based on the selected items. It also stores all purchased products in a cart for bill generation. The billing system reduces manual calculation errors and saves time during billing. The project uses concepts like classes, dictionaries, lists, loops, and conditional statements. It provides a simple and user-friendly menu-driven interface for easy operation. The generated bill displays product details, quantity, price, and total amount clearly. This project is useful for beginners to understand real-time applications of Python programming.

Introduction

The Billing System Project is a simple application developed using Python programming language. The main purpose of this project is to automate the billing process in shops and supermarkets. In traditional billing methods, manual calculations may take more time and lead to errors. This system helps in generating bills quickly and accurately. The project allows users to select products and enter the quantity of items purchased. It automatically calculates the total bill amount based on the selected products. The system also stores product details such as name, price, and quantity in a cart. Object-Oriented Programming concepts like classes and objects are used to make the project organized and efficient. The project includes loops, conditional statements, dictionaries, and lists for better functionality. This Billing System Project is useful for understanding real-world applications of Python programming and improving coding skills.

Concepts Used

- Python Programming
- Object-Oriented Programming (OOP)
- Classes and Objects
- Constructor (`__init__`)
- Dictionaries
- Lists
- Loops
- Conditional Statements
- Functions/Methods
- User Input Handling
- Menu-Driven Programming
- Data Storage
- Arithmetic Operations
- String Handling
- Iteration Techniques
- Bill Calculation Logic

Key Features

- Class and Object Creation
- Constructor (`__init__`)
- Dictionary for Product Storage
- List for Cart Management
- Loops (`while`, `for`)
- Conditional Statements (`if`, `elif`, `else`)
- User Input Handling (`input()`)
- Bill Generation System
- Total Amount Calculation
- Menu-Driven Program
- Product Selection by ID
- Quantity-Based Price Calculation
- Data Storage and Retrieval
- Simple and User-Friendly Interface
- Dynamic Cart Updating

Source Code

Project Name: Billing System

```
1 # =====
2 # BILLING SYSTEM PROJECT
3 # ----- CLASS -----
4 class BillingSystem:
5     # Constructor
6     def __init__(self):
7         # Dictionary
8         self.products = {
9             1: {"name": "Rice", "price": 50},
10            2: {"name": "Sugar", "price": 40},
11            3: {"name": "Milk", "price": 30},
12            4: {"name": "Oil", "price": 120},
13            5: {"name": "Biscuits", "price": 20}
14        }
15        # List
16        self.cart = []
17        self.total = 0
18
19    # ----- DISPLAY PRODUCTS -----
20    def display_products(self):
21        print("\n===== PRODUCT LIST =====")
22        # Loop
23        for pid, details in self.products.items():
24            print(pid, details["name"], "- Rs", details["price"])
25
26    # ----- ADD PRODUCT -----
27    def add_product(self):
28        while True:
29            self.display_products()
30            product_id = int(input("\nEnter Product ID : "))
31            # Conditional Statement
32            if product_id in self.products:
33                quantity = int(input("Enter Quantity : "))
34                product = self.products[product_id]
35                amount = product["price"] * quantity
36                # List Append
37                self.cart.append({
38                    "name": product["name"],
39                    "price": product["price"],
40                    "quantity": quantity,
41                    "amount": amount
42                })
43                self.total += amount
44                print("\nProduct Added Successfully")
45            else:
46                print("\nInvalid Product ID")
47                choice = input("\nDo You Want To Add More Products? (yes/no) : ")
48
49    # ----- DISPLAY BILL -----
50    def generate_bill(self):
51        print("\n===== BILL =====")
52        if len(self.cart) == 0:
53            print("Cart Is Empty")
54        else:
55            for item in self.cart:
56                print(item["name"], "| Price :", item["price"], "| Quantity :",
57                    item["quantity"], "| Amount :", item["amount"])
58            print("\n-----")
59            print("Total Bill : Rs", self.total)
60            print("-----")
61
62    # ----- MAIN MENU -----
63    def menu(self):
64        while True:
65            print("\n===== BILLING SYSTEM =====")
66            print("1. Add Products")
67            print("2. Generate Bill")
68            print("3. Exit")
69            choice = input("Enter Choice : ")
70            # Conditional Statements
71            if choice == "1":
72                self.add_product()
73            elif choice == "2":
74                self.generate_bill()
75            elif choice == "3":
76                print("\nThank You")
77                break
78            else:
79                print("\nInvalid Choice")
80
81    # ----- OBJECT CREATION -----
82    bill = BillingSystem()
83
84    # ----- METHOD CALL -----
85    bill.menu()
```

Output:

```
===== BILLING SYSTEM =====
1. Add Products
2. Generate Bill
3. Exit
Enter Choice : 1

===== PRODUCT LIST =====
1 Rice - Rs 50
2 Sugar - Rs 40
3 Milk - Rs 30
4 Oil - Rs 120
5 Biscuits - Rs 20

Enter Product ID : 2
Enter Quantity : 3

Product Added Successfully

Do You Want To Add More Products? (yes/no) : yes

===== PRODUCT LIST =====
1 Rice - Rs 50
2 Sugar - Rs 40
3 Milk - Rs 30
4 Oil - Rs 120
5 Biscuits - Rs 20

Enter Product ID : 5
Enter Quantity : 4

Product Added Successfully

Do You Want To Add More Products? (yes/no) : no

===== BILLING SYSTEM =====
1. Add Products
2. Generate Bill
3. Exit
Enter Choice : 2

===== BILL =====
Sugar | Price : 40 | Quantity : 3 | Amount : 120
Biscuits | Price : 20 | Quantity : 4 | Amount : 80

-----
Total Bill : Rs 200
-----

===== BILLING SYSTEM =====
1. Add Products
2. Generate Bill
3. Exit
Enter Choice : 3

Thank You
PS C:\Users\Akshitha\OneDrive\Desktop\mini project>
```

Conclusion

The Billing System Project is a useful application developed using Python programming language and Object-Oriented Programming concepts. This project helps in simplifying the billing process in shops and supermarkets. It allows users to select products, enter quantities, and generate bills easily. The system automatically calculates the total amount accurately and quickly. It reduces manual work and minimizes calculation errors during billing. The project uses important Python concepts such as classes, objects, loops, dictionaries, lists, and conditional statements. The menu-driven interface makes the system simple and user-friendly. This project improves programming knowledge and practical implementation skills. It also provides a better understanding of real-time software applications. Overall, the Billing System Project is an efficient and beginner-friendly project for learning Python programming.



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